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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

III / V - 2025 - 2026 Odd Semester

23AD410 - DATA EXPLORATION AND VISUALIZATION

HOTS (Higher Order Thinking Skills) Questions

UNIT I – EXPLORATORY DATA ANALYSIS

1.Why is EDA considered a critical step before applying machine learning models?

2.How does data transformation influence the insights derived from a dataset?

3.Compare classical statistical analysis and Bayesian analysis in decision making.

4.How can poor data aggregation lead to misleading conclusions?

5.Explain how pivot tables can be used for multidimensional analysis.

6.Design an EDA strategy for a real-world dataset and justify your steps.

7.Analyze how missing data impacts interpretation and suggest solutions.

8.Evaluate different data cleaning techniques for noisy datasets.

9.Propose a framework to merge multiple datasets effectively.

10.Justify the importance of visualization in EDA.

UNIT II – VISUALIZATION USING MATPLOTLIB

1 Why is visualization more effective than raw data interpretation?

2 Compare scatter plots and line plots for trend analysis.

3 How does color selection affect data perception?

4 Explain how histograms help in understanding distribution.

5 Evaluate the role of annotations in visualization.

6 Design a visualization dashboard using Matplotlib.

7Analyze the impact of 3D visualization on decision making.

8 Compare Matplotlib and Seaborn in terms of usability.

9 Propose visualization techniques for large datasets.

10 Justify the need for customization in data visualization.

UNIT III – UNIVARIATE ANALYSIS

1 How do measures of central tendency influence data interpretation?

2 Why is scaling important in univariate analysis?

3 Compare normalization and standardization.

4 How can inequality measures reveal hidden patterns?

5 Explain smoothing techniques in time series.

6 Design a method to analyze skewed data.
7 Analyze how outliers affect statistical measures.

8 Evaluate different distribution types in real datasets.

9 Propose techniques to handle variance in data.

10 Justify the importance of spread measures.

UNIT IV – BIVARIATE ANALYSIS

1 How do relationships between variables impact predictions?

2 Compare correlation and causation with examples.

3 Explain contingency tables in decision making.

4 How can scatter plots reveal hidden patterns?

5 Evaluate resistant lines in regression analysis.

6 Design a model to analyze two-variable datasets.

7 Analyze the effect of transformations on relationships.

8 Compare different visualization techniques for bivariate data.

9 Propose a method to detect anomalies in paired data.

10 Justify the importance of percentage tables.

UNIT V – MULTIVARIATE AND TIME SERIES ANALYSIS

1 How does introducing a third variable improve analysis?

2 Explain causal relationships in multivariate analysis.

3 Compare multivariate and bi-variate analysis.

4 How can time series data be cleaned effectively?

5 Evaluate resampling techniques in time series.

6 Design a model for healthcare data analysis.

7 Analyze trends using time-based indexing.

8 Compare longitudinal and cross-sectional data.

9 Propose visualization techniques for multivariate datasets.

10 Justify the importance of time series forecasting.